

Reinforcement Learning-Based Predictive Control for Power Electronic Converters

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Abstract—Finite-set model predictive control (FS-MPC) appears to be a promising and effective control method for power electronic converters. Conventional FS-MPC suffers from the time-consuming process of weighting factor selection, which significantly impacts control performance. Another ongoing challenge of FS-MPC is its dependence on the prediction model for desirable control performance. To overcome the above issues, we propose to apply reinforcement learning (RL) to FS-MPC for power converters. The RL algorithm is first employed for the automatic weighting factor design of the FS-MPC, aiming to minimize the total harmonic distortion (THD) or reduce the average switching frequency. Furthermore, by formulating the incentive for the RL agent with the cost function of the predictive algorithm, the agent learns autonomously to find the optimal switching policy for the power converter by imitating the predictive controller without prior knowledge of the system model. Finally, a deployment framework that allows for experimental validation of the proposed RL-based methods on a practical FS-MPC regulated stand-alone converter configuration is presented. Two exemplary control objectives are demonstrated to show the effectiveness of the proposed RL-aided weighting factor tuning method. Moreover, the results show a good match between the model-free RL-based controller and the FS-MPC performance.

Index Terms—Finite-set model predictive control (FS-MPC), model-free controller, reinforcement learning (RL), voltage source converter (VSC), weighting factor design.

I. INTRODUCTION

POWER electronic converters are one of the critical enabling technologies in dc microgrids, facilitating the integration of renewable energy sources, energy storage systems, electric vehicles, and ac systems [1]. Conventional linear control structures employ cascaded linear loops to control the converters with simple and analytical control synthesis. However,

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the slow transient performance, lack of flexibility for multiobjective controls, etc., limit its applications. Different advanced control methods have been proposed to address the obstacles of conventional cascaded linear control structures [2], among which model predictive control (MPC) stands out due to its simple inclusion of control objectives and straightforward design.

In general, MPC relies on the converter model to predict the future behavior of output voltage for all the possible switching configurations, and a cost function is used as the criterion for selecting the optimal switching state of the following sampling interval to achieve desired control performances [3]. Despite its advantages, there are still challenges with the MPC for converter control. The first is weighting factor design in the cost function for desirable control performance and balancing different control objectives. Conventionally, time-consuming simulations for optimal weighting factor selection are conducted by manual trial-and-error with different combinations of the parameters [4]. One of the pioneering proposals to relieve this is the branch-and-bound search method [5], while it is still too empirical. Different artificial intelligence techniques have also been applied to weighting factor design [6], [7], [8], [9], [10], which are generally categorized into metaheuristic optimization algorithm (MOA), fuzzy logic (FL), and artificial neural network (ANN) approaches.

MOA methods can find the weighting factors via multiobjective optimization [7], [8]. However, these methods strongly depend on system parameters, and the convergence speed is slow. Moreover, MOAs are not suitable for online implementation due to their high computational burden [6]. FL-based methods can also be employed for weighting factor design without manual selection [9]. However, this method cannot explicitly achieve weighting factor design based on various predefined desired system behaviors, and the design of the membership function is complex. The ANN approach achieves automatic weighting factor design for finite-set model predictive control (FS-MPC) [11]. A similar concept is also applied in [10], [12]. However, those methods require generating data from a simulation model or experimental setup under the parameter sweeping of the weighting factors. Despite their automatic design approach, an additional optimization stage is necessary to find the optimal weighting factors based on a user-defined fitness function. In [13], a model-free brain emotional learning (BEL) tuning method is proposed. However, the method introduces more coefficients required for tuning within the BEL method, increasing the complexity of controller design instead.

Another challenge is the high reliance on the converter model for the predictive controller to achieve optimized performance [14] and the high computation burden for online implementation, especially when the model order is high, e.g., multicell or multilevel converters. Even though different efforts have been carried out to mitigate these with the data-based methods, e.g., supervised neural network (NN)-based controller for voltage source converter (VSC) [15], [16], machine learning (ML)-based controller for modular multilevel converter (MMC) [17], and five-level flying capacitor converter [18], in principle, those methods require data generated by offline simulations based on the converter models in a supervised way to train the NN. To solve this, in [19], [20], a model-free data-driven predictive current control framework is proposed by synthesizing a model-free adaptive control (MFAC) with the FS-MPC for MMC and a three-level neutral-point-clamped converter. A nonlinear function is used to map the discrete-time dynamics of the controlled system. However, it requires historical operational data, and optimal performance requires extensive tuning of the introduced parameters.

Recently, reinforcement learning (RL) has gained much attention for applications in the power electronics field due to its model-free and self-learning characteristics. In [21], the RL is employed to tune the weights of the neuro-fuzzy controller for the active and reactive power control of distributed generators in microgrids. In [22], a self-adaptive active disturbance rejection control with RL-based controller coefficients tuning method is proposed to mitigate the instability issues caused by constant power load for dc/dc converters. Similarly, Cui et al. [23] proposed generating the duty ratio with RL to stabilize the dc/dc converter. In addition, different RL agents are applied to a dual active bridge converter to find the optimal phase shift angle of the triple phase shift modulation to minimize the reactive power [24], power losses [25], and current stress [26]. In [27], RL is integrated into the predictive control framework to improve the robustness of the control to different operating conditions by generating an optimal feedback control reference for the predictive controller.

Motivated by the advancements of RL, this article proposes to leverage the RL algorithms to reinforce FS-MPC for power converters in the mentioned two main aspects, i.e., facilitating weighting factors design process and mitigation of model dependence. It is noteworthy that the proposed methods can also be applied to other conventional controllers under certain application scenarios. Particularly, an RL-based automatic weighting factor design method is proposed for the FS-MPC. By defining different objectives, the RL agent will learn autonomously by interacting with the system to find the optimal weighting factor tuning policy for FS-MPC to achieve the desired control performance. It directly outputs the optimal weighting factors without an additional optimization stage. Furthermore, an RL-based controller is proposed to emulate the conventional FS-MPC for power converters. By properly configuring the RL agent, it can find the optimal switching strategy to achieve the desired control performance without any prior knowledge of mathematical models and parameters from the converter system. Eventually, for experimental verification of the proposed

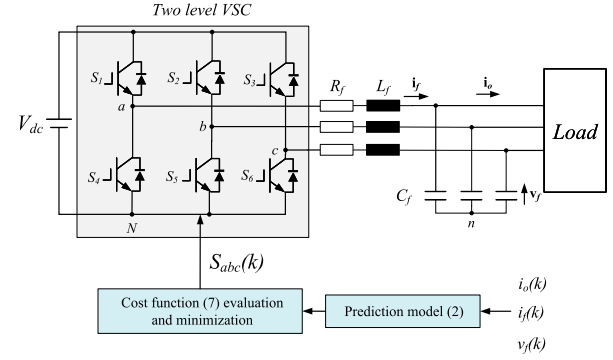


Fig. 1. Predictive control for VSC.

methods, a framework to implement the RL-based methods in a practical setup is demonstrated. The agents are trained offline and implemented online, avoiding time-consuming and storage-intensive online computation during training. The main contributions of the article are summarized as follows.

- 1) Automatic weighting factor design. An RL-based weighting factor design method is proposed to automate the weighting factors tuning process for FS-MPC to achieve optimal control regarding different control objectives.
- 2) Model-free RL control. An RL-based controller exhibiting comparable control performance to FS-MPC without prior mathematical models and parameters is developed.
- 3) A practical demonstration framework for transferring online RL from simulation to practical experimental setup.

The rest of the article is structured as follows. Section II introduces the model of the studied MPC-regulated system. Section III introduces the proposed RL applications in MPC controller design and imitation control. Section IV presents the implementation framework and experimental results for different case studies. Section V concludes the article.

II. SYSTEM MODEL

A. Converter Model

In Fig. 1, a two-level VSC interfaces between the load and dc source, where the FS-MPC is used to regulate it for the uninterruptible power supply (UPS) application. The control algorithm needs to maintain a smooth sinusoidal output voltage and give a fast response under load variations. The FS-MPC predicts converter behaviors in the future for all the possible switching combinations based on the prediction model, starting from the most recent measurements. A cost function denoting the desired control objective evaluates the performance of those predictions. The optimal switching policy for current states, i.e., with the least cost function value, will be applied to the converter.

The VSC is modeled in a stationary $\alpha\beta$ reference frame. Thus, amplitude-invariant Clarke transformation is employed to convert all the three-phase filter current $\mathbf{i}_f$, load current $\mathbf{i}_o$, and output capacitor voltage $\mathbf{v}_f$ into the corresponding $\alpha\beta$ frame from abc frame. In addition, the upper arm switching states $S = \{S_a, S_b, S_c\}$ for the three-phase legs are modeled

TABLE I
SWITCHING STATES AND VOLTAGE VECTORS

S_a	S_b	S_c	Voltage Vector $\mathbf{v}_i$
0	0	0	$\mathbf{v}_0 = 0$
1	0	0	$\mathbf{v}_1 = \frac{2}{3}v_{dc}$
1	1	0	$\mathbf{v}_2 = \frac{1}{3}v_{dc} + j\frac{\sqrt{3}}{3}v_{dc}$
0	1	0	$\mathbf{v}_3 = -\frac{1}{3}v_{dc} + j\frac{\sqrt{3}}{3}v_{dc}$
0	1	1	$\mathbf{v}_4 = -\frac{2}{3}v_{dc}$
0	0	1	$\mathbf{v}_5 = -\frac{1}{3}v_{dc} - j\frac{\sqrt{3}}{3}v_{dc}$
1	0	1	$\mathbf{v}_6 = \frac{1}{3}v_{dc} - j\frac{\sqrt{3}}{3}v_{dc}$
1	1	1	$\mathbf{v}_7 = 0$

as “ON” for “1” and “OFF” for “0”. Therefore, considering all the possible combinations of switching states for the three phases, there are eight switching configurations. The phase voltage of the VSC can be derived by multiplying dc link voltage and associated gating signal, as $v_{aN} = S_a \cdot v_{dc}$, $v_{bN} = S_b \cdot v_{dc}$, and $v_{cN} = S_c \cdot v_{dc}$. Therefore, the output voltage vector of the inverter can be expressed as [3]

$$\mathbf{v} = \frac{2}{3}(v_{aN} + \mathbf{a}v_{bN} + \mathbf{a}^2v_{cN}) \quad (1)$$

where $\mathbf{a} = e^{j(2\pi/3)}$, representing the 120° phase displacement between the phases.

Thus, the voltage vectors expressed in $\alpha\beta$ frame are listed in Table I.

The LC filter model can be expressed with the differential equations of the inductance current and capacitor voltage in state-space representation, as

$$\frac{d}{dt} \begin{bmatrix} \mathbf{i}_f \\ \mathbf{v}_f \end{bmatrix} = \mathbf{A} \begin{bmatrix} \mathbf{i}_f \\ \mathbf{v}_f \end{bmatrix} + \mathbf{B} \begin{bmatrix} \mathbf{v}_i \\ \mathbf{i}_o \end{bmatrix} \quad (2)$$

where $\mathbf{A} = \begin{bmatrix} -R_f/L_f & -1/L_f \\ -1/C_f & 0 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -1/L_f & 0 \\ 0 & -1/C_f \end{bmatrix}$.

For implementing the digital control, the model is discretized with a sampling time T_s based on the Euler forward discretization method [11].

B. Cost Function

Considering the computational complexity of the multistep prediction horizon, switching frequency, and topology of the two-level VSC, a single-step horizon is used for the case study in the article. The cost function evaluating the switching strategy is expressed as [14]

$$g = (v_\alpha^* - v_{f\alpha})^2 + (v_\beta^* - v_{f\beta})^2 \quad (3)$$

where v_α^* and v_β^* are the reference voltage.

An additional current reference term introduced to improve the steady-state performance is expressed as [28]

$$g_d = (C_f\omega_r v_\beta^* - i_{f\alpha} + i_{o\alpha})^2 + (C_f\omega_r v_\alpha^* + i_{f\beta} - i_{o\beta})^2. \quad (4)$$

Moreover, penalization terms to reduce the switching frequency and limit inductor current are employed, as follows:

$$sw = |\Delta S_a(i)| + |\Delta S_b(i)| + |\Delta S_c(i)| \quad (5)$$

$$h_{\lim} = \begin{cases} 0, & |\bar{i}_f| \leq i_{\max} \\ \infty, & |\bar{i}_f| > i_{\max} \end{cases} \quad (6)$$

where $|\Delta S_j(i)|$ ($j = a, b, c$) is 1 if switching states change in the phase leg at time i and 0 vice versa.

Therefore, the comprehensive cost function incorporating all the terms can be formulated as

$$g = (v_\alpha^* - v_{f\alpha})^2 + (v_\beta^* - v_{f\beta})^2 + \lambda_d g_d + \lambda_{sw} sw^2 + h_{\lim}. \quad (7)$$

It can be observed that two weighting factors assigned to different control objectives, i.e., λ_d and λ_{sw} , should be selected properly to achieve the desired control performance. Moreover, the FS-MPC relies heavily on the prediction model to predict the future behavior of the converter. How to mitigate the reliance on the system model is still an open research problem. In the following sections, the article proposes to employ the RL to solve the above issues.

III. MODEL-FREE REINFORCEMENT LEARNING METHODS FOR FS-MPC IN CONVERTER CONTROL

In this section, model-free reinforcement learning (MFRL) algorithms are applied to the FS-MPC for converter control. The application framework involves problem formulation, configuring the RL algorithms according to the case studies, and implementation. We first present preliminaries of RL algorithms. Then the proposed RL-based automated weighting factor design is presented for expected control performance. Finally, we propose RL-based imitation learning to achieve optimal performance and fast online implementation without model information.

A. Formulation of RL

RL paradigm is formulated as a Markov Decision Process (MDP) characterized with the tuple $\langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$ for the agent, where $\mathcal{S}$ is the state space, $\mathcal{A}$ represents the action space, P denotes the state transition function for characterizing the distribution over states regarding $s_t \rightarrow s_{t+1}$, R is the reward function for evaluating the goal of the problem, and $\gamma \in [0, 1]$ is the discount factor for balancing the agent learning with immediate and long-term reward. As shown in Fig. 2, the RL agent interacts with the environment iteratively. At each time step, the RL agent receives the environment state, $s_t \in \mathcal{S}$, and on that basis selects an action, $a_t \in \mathcal{A}$, according to the policy π . Then, at the time step $t + 1$, the agent receives a reward $r_t \in R$, and the environment proceeds to a new state s_{t+1} based on the transition function P . The iteration terminates until the RL agent finds the optimal policy as

$$\pi^* \in \operatorname{argmax}_{\pi} G(\pi) = \mathbb{E}_{\pi} \left[\sum_{k=0}^{\infty} \gamma^k r_t(s_t, a_t) \right] \quad (8)$$

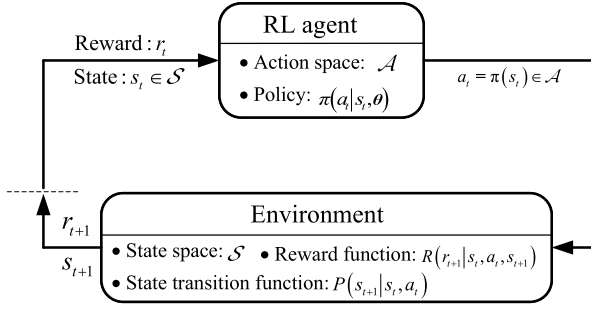


Fig. 2. Agent environment interaction in an MDP.

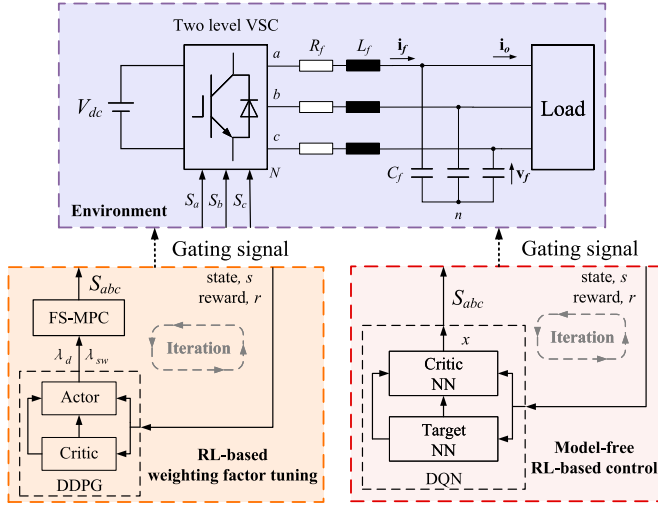


Fig. 3. Block diagram of the proposed RL-based framework for the converter system.

where $G(\pi)$ represents the accumulative reward over an infinite time horizon, and $\mathbb{E}[\cdot]$ denotes the expected value of a variable. The optimal policy guarantees that the expected return for all states and actions is greater than other policies.

There are two main RL algorithms, i.e., value-based and policy gradient algorithms. Value-based algorithms, e.g., Q -learning and its derivations, learn the Q function that evaluates the state-action pairs and take actions that maximize the Q -value. Policy gradient methods directly learn the policy $\pi(a_t|s_t, \theta)$, using either stochastic or deterministic estimates of policy gradient. Based on the characteristics of the formulated problem, the RL agents should be appropriately selected.

Deep deterministic policy gradient (DDPG) with continuous actions space and deep Q -learning network (DQN) with discrete action space are two basic paradigms of MFRL algorithms mentioned above, which are employed in the studies to demonstrate how the MFRL methods are applied to controller design and converter control respectively. The block diagram of the proposed RL-based framework for the converter system is shown in Fig. 3.

B. DDPG-Based Weighting Factor Design for FS-MPC

The left side of Fig. 3 presents the overall schematic of the proposed automatic weighting factor design approach. The environment consists of an MPC-regulated VSC for the UPS application. The measurements are selected as the states for the agent, along with a calculated reward based on the current status of the system. The RL agent will output the λ_d and λ_{sw} to the FS-MPC, and the system transits to a new state by executing the actions. Iteratively, the RL agent learns autonomously to find the optimal weighting factors tuning policy for FS-MPC. Eventually, the optimal weighting factors for desired control performance are obtained.

1) *RL Agents*: As mentioned in Section III-A, the weighting factor design problem is defined as a Markov decision process (MDP). The RL agent explores and exploits the action space to maximize the long-term accumulative reward over an infinite time horizon, which is expressed as

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}. \quad (9)$$

In order to find the optimal policy $\pi(a_t|s_t)$ to maximize the accumulative reward in (9), different recursive training algorithms can be applied. In the off-policy algorithms such as Q -learning, the action-value function $Q_\pi(s_t, a_t)$ is used to evaluate the expected return for action a_t with respect to the state s_t , which is expressed as Bellman equation below [29]

$$Q_\pi(s_t, a_t) \leftarrow Q_\pi(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_{a'} Q_\pi(s_{t+1}, a') - Q_\pi(s_t, a_t)] \quad (10)$$

where α is the learning rate of the RL agent.

However, the iterative approach cannot achieve desirable performance in high-dimensional real-world applications. For a more precise and efficient prediction of the Q value for each state-action pair, a deep neural network is employed [30]. In the DQN, transition sequences $e = (s_t, a_t, r_t, s_{t+1})$ are stored in a R -sized experience memory $\mathcal{M} = \{e_1, \dots, e_R\}$. During the training session, a minibatch of the tuples is randomly selected from $\mathcal{M}$ to train the NN via stochastic gradient descent. In addition, to enhance the stability and convergence of training Q -network with network parameter θ , a target network Q'_π is used. The target value is calculated as

$$y_t = r_t + \gamma \max_{a'} Q'_\pi(s_{t+1}, a'|\theta'_Q). \quad (11)$$

The loss function for updating the Q -network parameter is

$$L(\theta_Q) = \mathbb{E}[(y_t - Q_\pi(s_t, a_t|\theta_Q))^2]. \quad (12)$$

The target network keeps its separate network parameter θ' fixed for every T_{target} steps and is updated with the current weights of the Q -network as

$$\theta'_Q \leftarrow \tau \theta_Q + (1 - \tau) \theta'_Q \quad (13)$$

where the smoothing factor $\tau \ll 1$.

While DQN is feasible for problems with high-dimensional observation spaces, it can only handle a limited set of actions.

In the scenarios of weighting factor design, the weighting factor values cover a wide range within the action space, making the number of actions increase exponentially and the training intractable. Therefore, the RL agents capable of handling continuous action space using actor-critic networks, such as DDPG [31], are more suitable and employed. Similar to the DQN, the critic network is updated by minimizing the loss function in (12), another actor function $\mu(s|\theta_\mu)$ specifies the current policy by mapping states to the specific action. The actor-network is updated with the equation below [32]

$$\nabla_{\theta_\mu} J = \mathbb{E}_{s_t \sim \rho^\beta} [\nabla_{\theta_\mu} (s|\theta_\mu) | s=s_t \nabla_a Q_\pi(s, a|\theta_Q) | s=s_t, a=\mu(s_t)] \quad (14)$$

where $s \sim \rho^\beta$ represents the state s following the distribution β . Then the target networks of the critic and actor networks are updated with (13). In addition, a noise model $\mathcal{N}$ decaying by training episodes is usually used to help the agent balance exploration and exploitation better [31].

2) *State and Action Sets*: As shown in Fig. 3, the DDPG agent receives states from the MPC-controlled VSC system and tunes the weighting factors of the MPC controller. As the goal is to achieve the desired control performance, therefore, the state is defined as $s = [v_{f\alpha}, v_{f\beta}, e_{f\alpha}, e_{f\beta}, \text{THD}]$, where e_α and e_β are the reference voltage tracking error. The action set is defined as $a = [\lambda_d, \lambda_{sw}]$. In addition, for more efficient training, the action exploration space is limited within a feasible range [0, 10] for both actions.

3) *Reward Function Design*: The reward is formulated according to desired control performances, minimizing the total harmonic distortion (THD) and reducing the average switching frequency. Therefore, the reward function should be properly designed for a specific control objective. Two exemplary goals are studied, i.e., minimization of the THD exclusively and simultaneously decreasing the switching frequency f_{sw} .

For the first case study, the reward function is defined as

$$r = -\text{THD}^2. \quad (15)$$

Therefore, the reward will incentivize the RL agent to tune the two weighting factors of the FS-MPC to minimize the THD.

For the second case study, the reward function is defined as

$$r = -(k_{thd} \cdot \text{THD}^2 + k_{sw} \cdot f_{sw}^2) \quad (16)$$

where k_{thd} and k_{sw} are the coefficients for the THD and f_{sw} respectively. The f_{sw} is calculated by averaging the total SW from (5) within a certain operating time [11]. The coefficients could be selected according to the importance of the control objectives. In the case study, more importance is given to reducing the THD, thus $k_{thd} = 2$ and $k_{sw} = 1$. The training process of DDPG for FS-MPC controller design is described in Algorithm 1.

C. DQN-Based Self-Learning Imitation Controller of FS-MPC

As shown in the right side of Fig. 3, to address the reliance on the system model for conventional FS-MPC, we propose a novel model-free RL-based imitation controller to emulate

Algorithm 1: DDPG-Based Controller Design for FS-MPC

Input: $[v_{f\alpha}, v_{f\beta}, e_\alpha, e_\beta, \text{THD}]$

Output: Weighting factors λ_d and λ_{sw}

- 1: Initialize replay buffer $\mathcal{M}$
- 2: Initialize critic network Q and actor-network μ with random weights θ_Q and θ_μ
- 3: Initialize target network Q' and actor-network μ' with weights $\theta'_Q = \theta_Q$, $\theta'_\mu = \theta_\mu$
- 4: **for** episode = 1 to M **do**
- 5: Initialize a random noise $\mathcal{N}$ for action exploration
- 6: Receive initial observation at state s_1
- 7: **for** iteration = 1 to T **do**
- 8: Select action with current policy and noise
- 9: Excute the action a_t , observe the reward r_t based on (15) or (16) and new state s_{t+1}
- 10: Store tuple (s_t, a_t, r_t, s_{t+1}) in $\mathcal{M}$
- 11: Randomly sample the mini-batch e_j from $\mathcal{M}$
- 12: Set $y_j = r_j + \gamma Q'(s_{i+1}, \mu'(s_{i+1}|\theta'_\mu) | \theta'_Q)$
- 13: Update the critic by minimizing (12)
- 14: Update the actor with (14)
- 15: Update the target networks using (13)
- 16: **end for**
- 17: **end for**

the FS-MPC to achieve desired control performance for power electronic converters with light online computation requirement and without prior model information.

Fig. 3 presents the RL-based controller for VSC. Due to the limited discrete switching states listed in Table I, the DQN agent is more suitable for the case study. The DQN agent will observe the system states and receive the reward for the executed action, i.e., the voltage vector corresponding to the switching states x . By properly designing the reward function as incentives for training RL agents based on the cost function of FS-MPC, the RL agent will learn autonomously to find the optimal switching strategy for the converter to achieve desirable performance similar to the FS-MPC.

1) *State and Action Sets*: As shown in Fig. 3, the observed states include reference voltage (v_α^* , v_β^*), output voltage deviations ($\Delta v_{f\alpha}$, $\Delta v_{f\beta}$), the filter currents ($i_{f\alpha}$, $i_{f\beta}$) and the voltage vector x_{old} of previous sampling period. The action space consists of all the possible voltage vectors. Due to the repeated voltage vector for two switching states, the action space is defined as $x = [1:7]$, corresponding to the switching states.

2) *Reward Function Design*: It is noteworthy that different control objectives could be realized by fine-tuning the reward function, similar to the cost function design for FS-MPC. In this study, an exemplary control strategy for regulating ac voltage for VSC is presented. The reward function of RL should be finite for feasible and efficient training. Therefore, to mimic the optimal switching state selection strategy of FS-MPC, the reward function is formulated based on the single-step horizon cost function, expressed as

$$r = -[(v_\alpha^* - v_{f\alpha})^2 + (v_\beta^* - v_{f\beta})^2]. \quad (17)$$

Algorithm 2: DQN-Based Imitation Controller of FS-MPC for Power Converter

Input: $[v_\alpha^*, v_\beta^*, \Delta v_{f\alpha}, \Delta v_{f\beta}, i_{f\alpha}, i_{f\beta}, x_{old}]$
Output: Voltage vector number x

- 1: Initialize replay buffer $\mathcal{M}$ to capacity R
- 2: Initialize Q -network with random weights θ_Q
- 3: Initialize target Q -network with weights $\theta'_Q = \theta_Q$
- 4: **for** episode = 1 to M **do**
- 5: Receive initial observation at state s_1
- 6: **for** iteration = 1 to T **do**
- 7: With probability ϵ select a random action a_t
 otherwise select $a_t = \max_a Q_\pi(s_t, a|\theta)$
- 8: Execute the action a_t , observe the reward r_t
 calculated with (17) and state s_{t+1}
- 9: Store tuple (s_t, a_t, r_t, s_{t+1}) in the $\mathcal{M}$
- 10: Randomly select a mini-batch e_j from $\mathcal{M}$
- 11: Set $y_j = \begin{cases} r_j, & \text{if episode terminates at step } j+1 \\ r_j + \gamma \max_a Q'_\pi(s_{j+1}, a|\theta'_Q), & \text{otherwise} \end{cases}$
- 12: Perform gradient descent on (12) regarding the network parameter θ_Q
- 13: Every T_{target} steps reset $\theta'_Q = \theta_Q$
- 14: **end for**
- 15: **end for**

The training process is shown in **Algorithm 2**.

Regarding stability analysis, after the optimal control policy is learned using the proposed RL approach, a describing function method can be employed [33], which approximates the nonlinear MFRL-based loop with a quasi-linear representation. In this way, the dynamic characteristics of the MFRL imitation controller can be obtained.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the proposed RL-aided weighting factor design method of FS-MPC and the MFRL-based controller for power converters are validated experimentally. The experimental setup, with parameters listed in Table II, is shown in Fig. 4. It includes a two-level VSC consisting of three IMPERIX PEB 8024 H-bridge power modules, interfaced with a linear load via an LC filter. An IMPERIX B-Box RCP control platform is used to implement the proposed methods as in Fig. 5.

A. From Simulation to Practical Implementation of Online RL

The development of RL-based controller design and imitation controller can be split into two main steps. First, the RL agent is trained in MATLAB/SIMULINK with a model matching testbed parameters. As presented in Section III, by properly configuring the RL agents respectively for different case studies, the RL agents learn autonomously through interacting with the system. In particular, the DDPG agent tunes the optimal weighting factors of the FS-MPC for different control objectives, and the DQN agent emulates the optimal switching

TABLE II
EXPERIMENTAL TESTBED PARAMETERS

Parameter Sets	Parameter Values
DC link voltage	$v_{dc} = 520 \text{ V}$
Output filter	$L_f = 2.5 \text{ mH}, R_f = 0.013 \Omega, C_f = 30 \mu\text{F}$
Reference voltage	$V_r = 200 \text{ V}, f_r = 50 \text{ Hz}$
Load	$R_{load} = 50 \Omega$
Sampling time	$T_s = 50 \mu\text{s}$

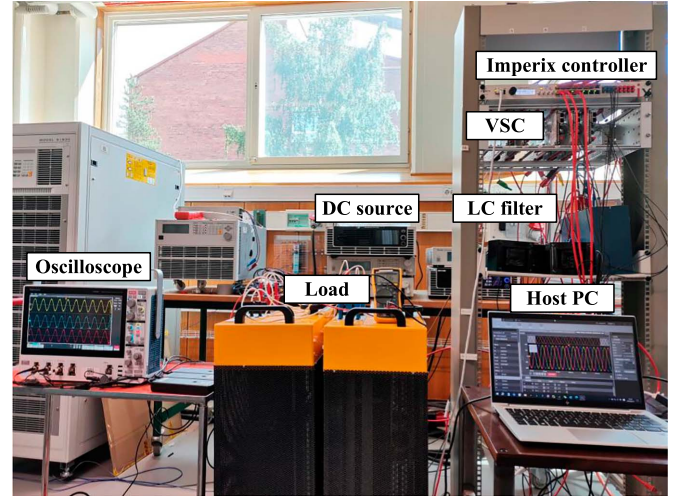


Fig. 4. Experimental setup of a two-level VSC system.

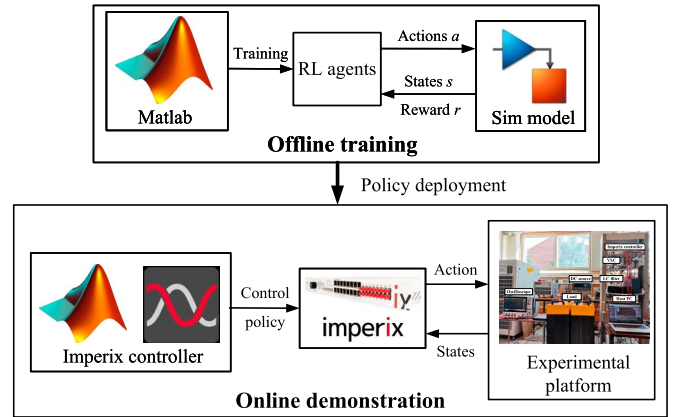


Fig. 5. Diagram of the RL-based methods development process from offline simulation to online demonstration.

strategy of conventional FS-MPC for VSC to achieve desired control performance without the prediction model. In this article, all the simulations have been performed in Matlab/Simulink with a PC Intel(R) Core(TM) i5-9500 CPU at 3.00GHz and 32 GB RAM.

Afterward, the optimal policies from the well-trained online RL agents are transferred from simulation to practical setup using the IMPERIX platform. The training on the devices with the simulation model can stay the same with the real power converter system. In the practical implementation process, the

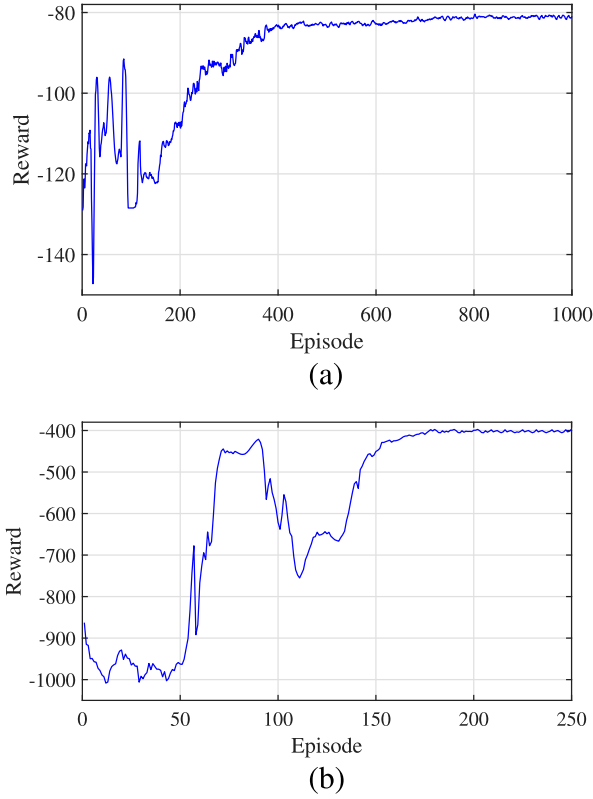


Fig. 6. Average accumulated reward during training process. (a) Scenario A. (b) Scenario B.

optimal policy is deployed by configuring the corresponding measurements as the input for the RL agents, where the actions for achieving optimal control performance are output. The workflow of the above is presented in Fig. 5.

B. DDPG-Based Weighting Factor Design for FS-MPC

Different case studies are conducted to verify the proposed RL-based weighting factor design method. The training results are shown in the figures as follows.

1) *Scenario A: Minimization of THD*: In this scenario, the reward in (15) is designed for training the agent, which encourages it to find the optimal weighting factors for minimization of the THD. The training process regarding the accumulated average reward over the training episodes is presented in Fig. 6(a). It shows that the RL agent keeps exploring the action space during the initial sessions, resulting in a fluctuation of the accumulated average reward. During episodes 180–400, the agent learns to converge to the optimal weighting factors by balancing the exploration and exploitation process, and the reward increases. After around 400 training episodes, the agents exploit and converge to the optimal policy, where the weighting factors for minimizing the THD are obtained.

The weighting factor design policy from the well-trained agent for THD minimization is deployed and applied with the demonstration framework, where the obtained optimal weighting factors are $\lambda_d = 1.38$, $\lambda_{sw} = 2$. The line-to-line capacitor voltage without and with the optimal weighting factors is shown

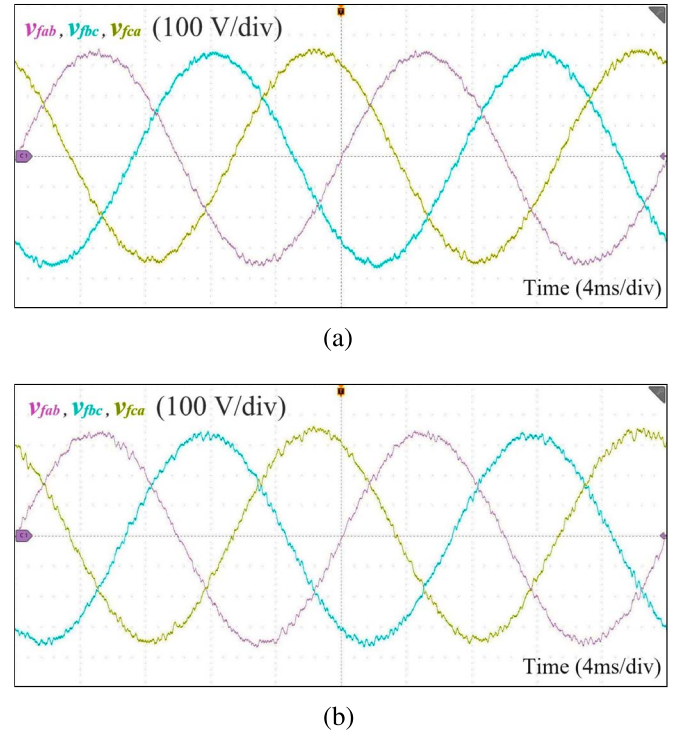


Fig. 7. Experimental capacitor voltage waveforms under different conditions with RL-based tuning method. (a) Scenario A: $\lambda_d = 1.38$, $\lambda_{sw} = 2$. THD = 2.22%, $f_{sw} = 4.55$ kHz. (b) Scenario B: $\lambda_d = 0.75$, $\lambda_{sw} = 7.04$. THD = 2.61%, $f_{sw} = 3.62$ kHz.

in Figs. 11(a) and 7(a), respectively. The calculated THD and average switching frequency is 2.22%, 4.55 kHz with the RL-tuned weighting factors compared with the 3.37%, 4.95 kHz of the weightless counterpart.

2) *Scenario B: Minimization of THD With Low f_{sw}* : To achieve the trade-off between the two control objectives in this case study, the RL agent learns to tune the weighting factors to minimize the THD with a low average switching frequency. The training process is presented in Fig. 6(b), where the agent also experiences the exploration, learning, and exploitation stages to find the optimal weighting factors.

The optimal weighting factors design policy for FS-MPC from the RL agent is deployed and validated experimentally, where the optimal weighting factors are $\lambda_d = 0.75$, $\lambda_{sw} = 7.04$. The results of the line-to-line capacitor voltage are shown in Fig. 7(b), and the THD and average switching frequency regarding the RL-tuned weighting factors is 2.61%, 3.62 kHz.

The above results verify that the control performance is improved based on different desired objectives with the RL-tuned weighting factors. Furthermore, 100% linear load step tests for both cases are conducted to illustrate the marginal influence of weighting factors on the system's transient performance, which are presented in Fig. 8. The voltage error in both case studies converges quickly when there is a load step, resulting in quick convergence of the root-mean-square (RMS) of voltage deviation within the limit. Thus, the voltage deviation complies with the limits indicated in the standard IEC 62040-3.

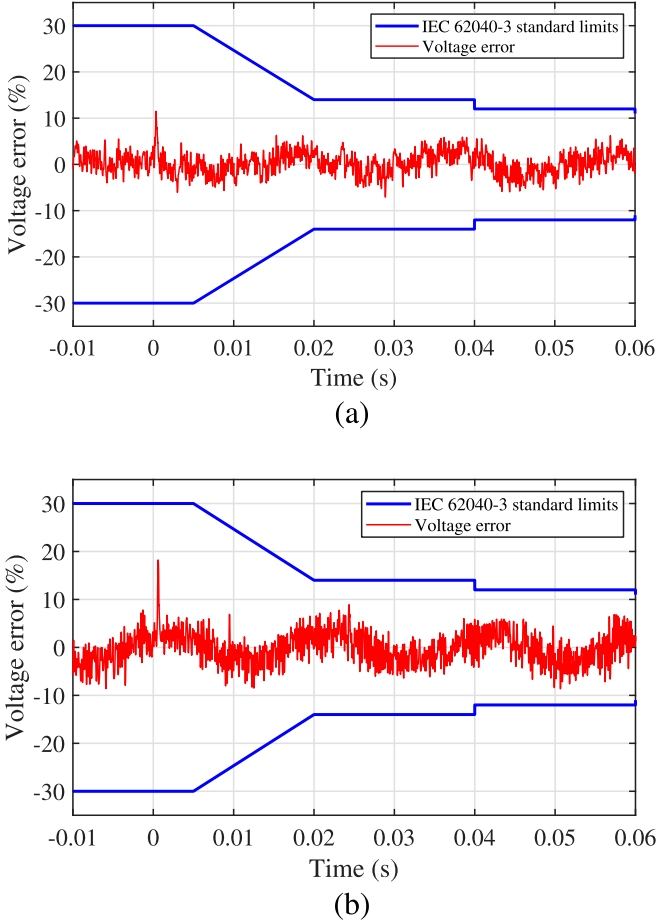


Fig. 8. Transient performance under 100% load step with the RL-tuned optimal weighting factors. (a) 100% load step with $\lambda_d = 1.38$, $\lambda_{sw} = 2$. (b) 100% load step with $\lambda_d = 0.75$, $\lambda_{sw} = 7.04$.

3) *Comparison: Benchmark ANN-Based Weighting Factor Design Approach:* Previous works in [11] proposed an ANN-based weighting factor design method, where data are generated by sweeping the possible value combinations of each weighting factor via simulations. Alternatively, for higher accuracy, experimental data with different weighting factor combinations could be used, which is more time-consuming and requires human monitoring. Thus, it can be used as a baseline benchmark to compare the selected values from the proposed RL-based weighting factor design method, further verifying effectiveness for different control objectives. It is noteworthy that the ANN-based method requires an additional optimization stage to optimize the weighting factors for different performance goals, thus it cannot be implemented online. In addition, the performance of the ANN-based method is heavily dependent on the generated data, which is also a time-consuming process to collect a sufficient amount. During the online demonstration, the implementation process can be simplified by obtaining the fine-tuned weighting factors during offline training and applying these factors directly to the FS-MPC in real time [6].

The step-by-step weighting factor design process can be referred to [11]. Specifically, a range of λ_d and λ_{sw} sweep

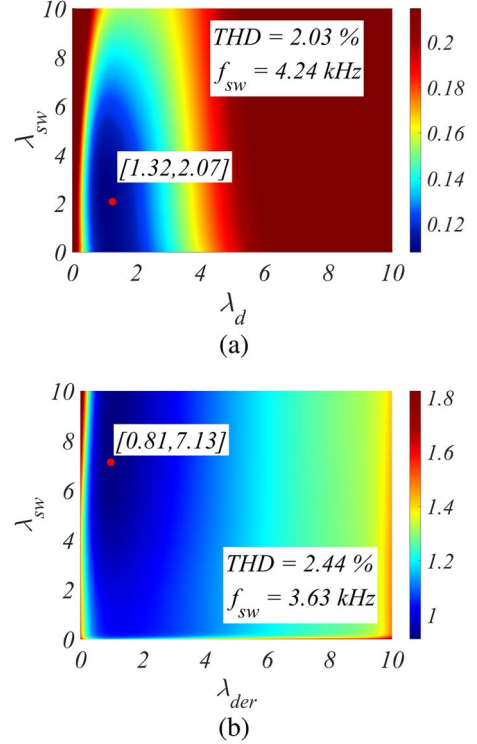


Fig. 9. ANN approach for weighting factor design. (a) Case study 1: minimizing THD. (b) Case study 2: minimizing THD with low f_{sw} .

between 0 and 10 was selected, with a step of 0.1 to cover the same design range. Therefore, a total of $101^2 = 10201$ parameter combinations are simulated, and the corresponding THD and average switching frequency are also obtained. The data is used to train an ANN, which is then used to optimize weighting factors for different design objectives. An exhaustive search algorithm is used to optimize the weighting factors by finding the minimum of the fitness functions.

The plots for the two case studies are shown in Fig. 9(a) and 9(b). For the first objective of minimization of THD, the best λ_d and λ_{sw} is around 1.32 and 2.07. However, it is noteworthy that the precise selection of values is not as significant, as the selection of λ_d and λ_{sw} covers a range that still yields performance close to the optimal. For the second case study, aimed at minimizing THD with low f_{sw} , the optimal weighting factor values are approximately 0.81 and 7.13 for λ_d and λ_{sw} , respectively.

Based on the above analysis, the RL-based approach can find the optimal weighting factors for various desired control performances. Moreover, compared with the ANN approach, the proposed method can be straightforwardly implemented online without an additional optimization stage.

C. DQN-Based Imitation Controller for VSC

The training process of the DQN agent learning the FS-MPC controller for VSC is shown in Fig. 10. At the initial training session, the agent explores the learning space without any prior system information by selecting the switching states randomly,

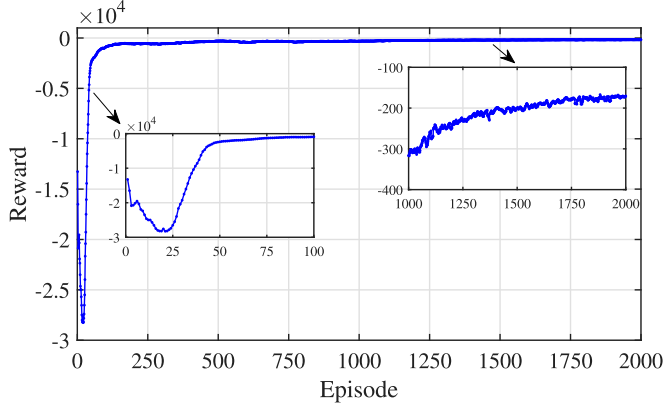


Fig. 10. Average accumulated reward during training process.

resulting in a sharp drop in average accumulated reward due to the significant deviation of regulated capacitor voltage from the reference voltage. Afterward, the DQN agent starts to balance the exploration and exploitation process, where the designed reward function helps the agent learn to select the optimal switching states based on the input measurements and previous switching actions. In this way, the accumulated average reward increases and converges gradually, indicating the DQN agent exploits and finds the optimal switching state selection policy, emulating the FS-MPC.

The optimal control policy from the well-trained DQN is validated experimentally using the deployment framework, and the results are presented in Fig. 11(b). The difference in THD and f_{sw} between the conventional FS-MPC without weighting factor and MFRL-based imitation controller are 0.12% and 0.37 kHz, respectively, confirming an effective learning of the FS-MPC control for VSC. More importantly, the DQN-based controller does not require any prior knowledge of the system and is trained autonomously without explicit labels, compared with conventional FS-MPC [14] and supervised ANN-based imitator [15] for power converter control.

To validate the DQN-based controller's transient performance, a reference voltage step is applied, as shown in Fig. 12(b). It shows that the RL-based controller achieves performance comparable to the conventional FS-MPC. In addition, the performance of the controllers is tested under a load step ($R_{load} = 100 \rightarrow 50 \Omega$). The transient performance under the load step is shown in Fig. 13. It can be observed that the RL-based controller achieves good tracking performance and fast transient response, similar to the conventional FS-MPC.

Moreover, despite load types, the model-free and self-learning characteristics of the RL agent enable it to learn the optimal control policy under different load conditions through interaction with the converter system. This allows for desirable control performance without prior knowledge of the system dynamics. The effectiveness of the proposed MFRL controller is validated under unknown unbalanced loads. Without prior knowledge of loads, results show that the proposed MFRL

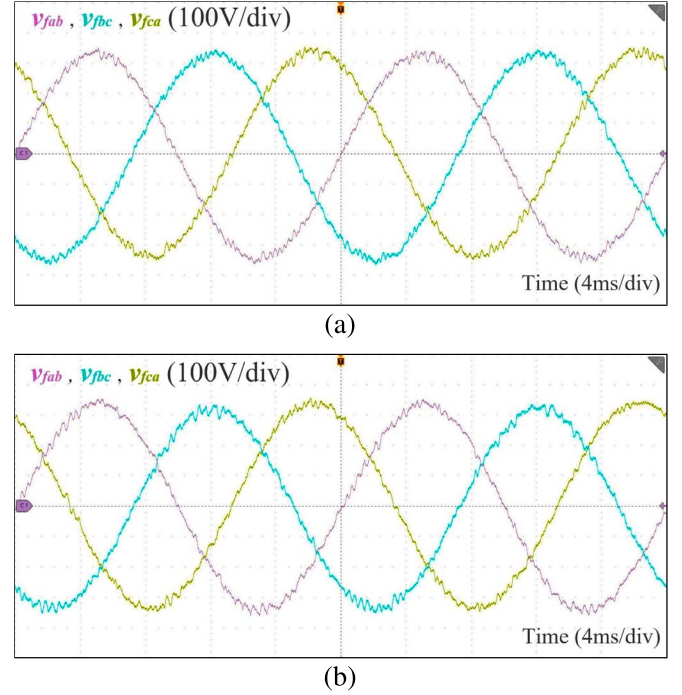


Fig. 11. Performance comparison of standard FS-MPC and MFRL-based controller. (a) Standard FS-MPC. THD = 3.37%, $f_{sw} = 4.95$ kHz. (b) MFRL controller. THD = 3.49%, $f_{sw} = 4.58$ kHz.

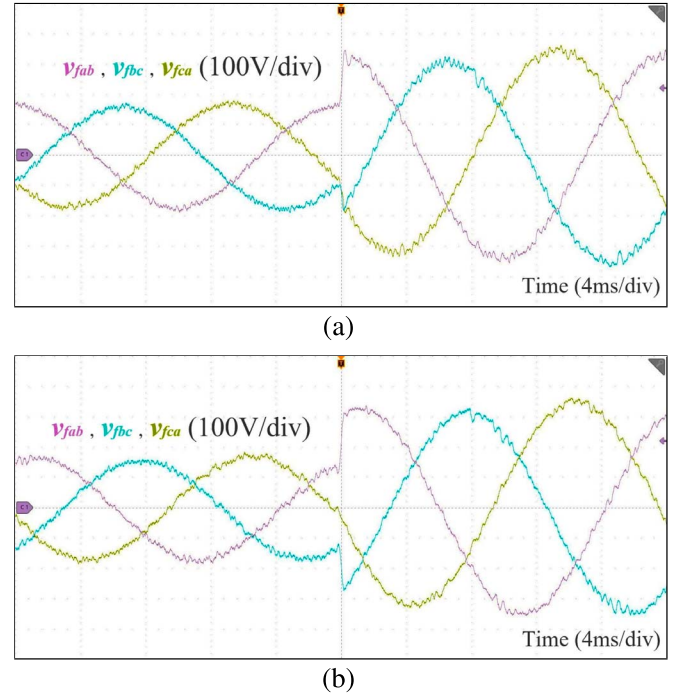


Fig. 12. Output voltage when applying reference voltage step (100 V → 200 V). (a) Reference voltage step for FS-MPC. (b) Reference voltage step for RL-based controller.

controller can still achieve desirable control performance. Despite the unbalanced load current shown in Fig. 14(b), the three-phase capacitor voltage is stable in amplitude and frequency, with a THD of 3.56%, as illustrated in Fig. 14(a).

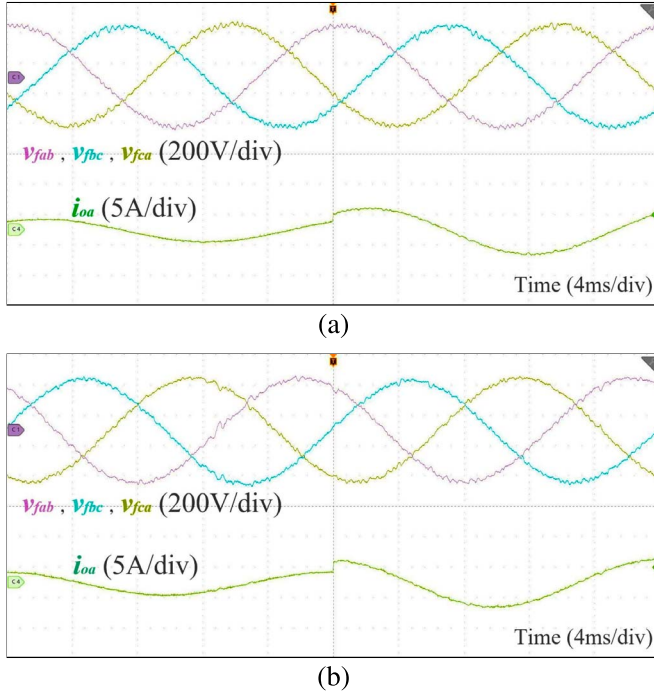


Fig. 13. Transient performance comparison under load step. ($R_{load} = 100 \rightarrow 50 \Omega$). (a) Conventional FS-MPC. (b) MFRL controller.

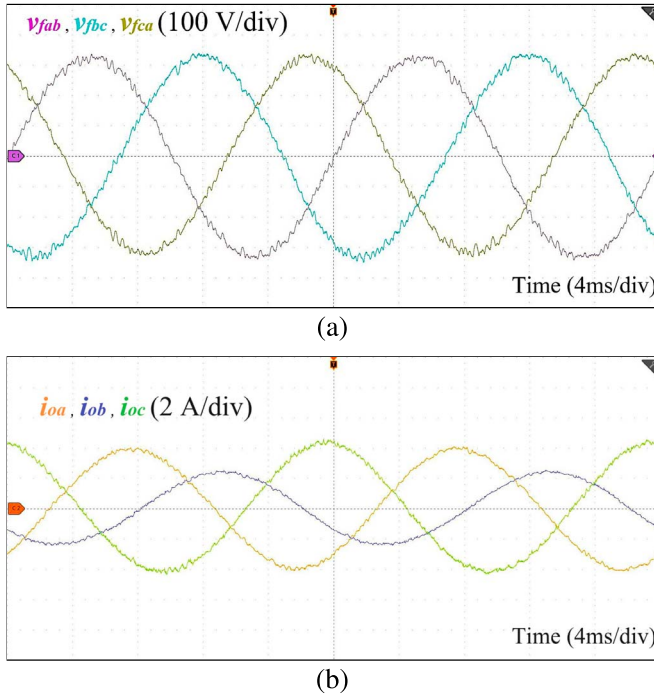


Fig. 14. Performance under unknown unbalanced loads. (a) Capacitor voltage. (b) Load current.

D. Computation Complexity

The computational complexity is crucial for implementing the proposed approaches in practical applications. The MPC controller has cubic complexity considering the need for matrix

TABLE III
CHARACTERISTICS COMPARISONS AMONG DIFFERENT CONTROL STRATEGIES

Methods	Advantages	Disadvantages
Linear control methods	Simple and analytical control design.	1) Lack of flexibility; 2) nonlinearity limitation; 3) limited dynamic performance.
FS-MPC [14]	1) Intuitive design; 2) handling nonlinearities and constraints; 3) fast dynamic response.	1) Weighting factor selection; 2) model dependency;
Data-driven methods [15], [19], [20]	1) Handling nonlinearities; 2) model-free characteristics.	1) Data dependency; 2) manual data processing.
Proposed method	1) Model-free characteristics; 2) handling nonlinearities; 3) computationally efficient online implementation.	Time-extensive offline training.

multiplication and inversion, especially for multicell or multi-level converters within the control horizon where the number of switching states increases. In comparison, the vast majority of the computing cost of the proposed RL-aided weighting factor design and MFRL-based imitation controller comes from the offline training stage at edge devices, and the time consumption is not critical. Once trained successfully, the RL-based methods involve only computationally light matrix manipulation within the neural networks, which enjoy linear computational complexity.

More importantly, with a simple deep neural network structure, the RL can map the nonlinear relationship between the input and output for most problems without constructing more hidden layers, thus not inducing more computational burden. There is potential for the proposed RL methods to solve more complex problems, e.g., more weighting factors and multilevel or multicell converter control. Therefore, despite the time-consuming training process, the practical implementation of the proposed RL-based control strategy is quite tractable.

To this end, Table III presents a comprehensive comparison of various control strategies for power converters. The proposed RL-based control method is compared with conventional linear control methods, FS-MPC [14], and data-driven controllers [15], [19], [20]. The proposed method inherits its good control performance of FS-MPC, as demonstrated in the experimental results. However, it presents a conceptually different, model-free approach that autonomously learns the optimal control strategy for the power converter without requiring prior knowledge of mathematical models and parameters of the converter system. Additionally, the neural network of the RL agent not only enables it to handle system nonlinearities effectively but also ensures computational efficiency in real-time implementation.

V. CONCLUSION

This article provides a guideline for applying the RL to the controller design and self-learning controller for power converters, starting from the problem formulation and RL agent type selection to the reward design and policy deployment in a practical converter testbed. In particular, a DDPG-based automatic weighting factor tuning method and a DQN-based imitation controller emulating the optimal switching states selection of FS-MPC are proposed. The proposed methods are model-free and thus require no prior knowledge of the system, improving or inheriting the good performance of conventional FS-MPC. In addition, with the proposed deployment framework for transferring the online RL agents from simulation to practical implementation, the proposed methods are validated experimentally. The results show that the RL-aided weighting factor design for FS-MPC achieves the desired control performance. In addition, the results also confirm that the RL-based imitation controller achieves a comparable control performance to conventional FS-MPC for VSC.

Findings in this article open interesting topics for the future development of the proposed method. The RL-based automatic controller design method is also applicable to tuning the coefficients of other controllers. In addition, due to the inherent nonlinear mapping capability of employed deep neural networks in RL, the proposed MFRL-based imitation controller may still work well without adding hidden layers. Future research could explore the possibilities of the imitation controller for algorithms for multilevel converters or modular multilevel topologies, where the computational burden increases exponentially due to the increasing number of switching states. In addition, future work can extend to a physics-informed approach to ensure system stability such as using the Lyapunov function to shape the reward to guide or constrain RL agent within the stability region during the learning process.

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